

Deferred Validity And Generalization Backlog

This document tracks the research-limit reductions that are still worth doing after the project is stable and the current curriculum results are acceptable.

This document is separate from RESEARCH_CHECKLIST.md. The checklist defines the minimum evidence bar for the active study. This document covers the follow-up work needed to make the claims narrower, stronger, and easier to defend under review.

How To Use This Backlog

- Treat these items as deferred work, not blockers for the current rerun campaign.
- Keep the corresponding threats-to-validity language in the reports until the relevant checklist is completed.
- When a section is completed, update both this document and the report wording that depends on it.

1. Validate Behavioral Heuristics

Goal: show that the project metrics for looping, plateauing, exploration, and pressure response track meaningful behavior rather than arbitrary proxy movement.

- [] Create a labeling rubric for looping, plateauing, exploration spike, same-opponent recovery, overfitting, and degradation.
- [] Sample a stratified set of epochs across curriculum families, replicates, and outcomes.
- [] Produce a blinded annotation packet so human reviewers do not see the condition name or hypothesis.
- [] Have at least two annotators label the packet independently.
- [] Measure inter-annotator agreement.
- [] Compare automated metrics against the human labels.
- [] Revise or rename metrics that do not align well with the annotations.
- [] Add a report section summarizing metric-validation results.

Why this matters:

- Current curriculum metrics are still heuristic operationalizations.
- Reviewers will reasonably ask whether the automated signals correspond to real strategy changes.

2. Separate Training-Time Selection From Final Evaluation

Goal: make it clear that replay checks and holdout spot checks are selection tools, not the final evidence of robustness or generalization.

- [] Freeze a held-out benchmark panel before analyzing new runs.
- [] Expand the held-out panel so it covers multiple opponent archetypes and map regimes.
- [] Run final accepted learner policies against the full held-out panel, not just selection-time spot checks.
- [] Ensure evaluation uses fixed seeds and no online adaptation.
- [] Report training metrics and held-out evaluation metrics in separate sections.
- [] Prevent aggregate conclusions from collapsing training success and evaluation success into one sentence.

- ☐ Add explicit report wording that selection-time replay checks are inner-loop filters.

Why this matters:

- Training-time robustness probes improve selection discipline, but they do not replace a real test set.

3. Strengthen Replication And Statistical Discipline

Goal: reduce the chance that reported effects are artifacts of one seed, one run family, or one noisy comparison.

The current operational path for this work is docs/PHASE_4_CAUSAL_TRANSFER_PROTOCOL.md plus analyze_causal_transfer.py.

- ☐ Run at least three replicated live runs for every family used in claims.
- ☐ Prefer five replicates for the strongest behavioral claims if cost allows.
- ☐ Freeze the primary metrics before inspecting results.
- ☐ Report uncertainty intervals for the main learner-centric outcomes.
- ☐ Report effect sizes, not only directional differences.
- ☐ Keep learner-centric summaries separate from opponent-role summaries.
- ☐ Add a short analysis note describing which claims are primary and which are exploratory.

Why this matters:

- Long-horizon adaptive systems are variable, and reviewers will expect replication discipline similar to RL best practice.

4. Add Structured Qualitative Review

Goal: support quantitative claims about novelty, recovery, churn, or overfitting with concrete case evidence.

- ☐ Define a fixed process for selecting notable epochs across runs.
- ☐ Include prompts, submitted code, executed-code traces, opponent context, and path artifacts for each selected epoch.
- ☐ Identify at least one example each of:
 - ☐ escape from a losing regime
 - ☐ churn without improvement
 - ☐ replay failure or archive-induced recovery
 - ☐ holdout failure after strong training performance
- ☐ Add qualitative appendices or companion notes for those cases.
- ☐ Use those cases to check whether the automated metrics agree with the manual evidence.

Why this matters:

- Code novelty alone is not enough to justify words like innovation or new strategy class.

5. Broaden Generalization Claims Carefully

Goal: move from "works on this benchmark family" toward stronger and more defensible generalization statements.

- ☐ Split held-out evaluation into near-OOD and far-OOD regimes.

- [] Add variation in board size, obstacle density, resource layout, and spawn rules.
- [] Test unseen opponent families that were never present during training.
- [] Report generalization separately by regime instead of averaging everything together.
- [] Avoid broad generalization claims even after the added routing benchmark families are run with replicated official suites. The completed 20-offset TSP/ATSP/CVRP pass is mixed and family-dependent, so broader replay-aware claims still need tighter scope and better mechanism-level explanation.
- [] If feasible, add a second related environment and rerun the evaluation protocol there.

Why this matters:

- Generalization is usually task-family-specific unless the benchmark scope is widened explicitly.

6. Improve Quality-Diversity Interpretation

Goal: make diversity-aware selection claims more rigorous and less dependent on lexical novelty.

The current operational path for this work is docs/PHASE_4_CAUSAL_TRANSFER_PROTOCOL.md plus `analyze_causal_transfer.py`.

- [] Compare lexical novelty and behavioral novelty directly across accepted and rejected candidates.
- [] Check whether behavior-cell coverage predicts held-out robustness better than code novelty.
- [] Evaluate whether elite-archive coverage grows in a stable way across replicates.
- [] Add plots showing performance versus behavior-cell coverage over training.
- [] Identify whether novelty-aware selection finds genuinely different successful policies or mainly filters churn.

Why this matters:

- The current curriculum results suggest that selection is helping more by rejecting poor churn than by rewarding many successful novel solutions.

7. Report-Language Tightening

Goal: keep the public-facing reports conservative and researcher-appropriate until the stronger evidence exists.

- [] Keep terms like heuristic, proxy, suggests, and preliminary where warranted.
- [] Avoid using innovation as a headline claim without behavioral validation.
- [] Avoid using generalization without held-out evaluation evidence.
- [] Avoid using cheating when the evidence is only a rule-boundary indicator.
- [] Update the reports when metric validation or broader evaluation is complete.

Why this matters:

- Strong claims are much harder to defend than careful, scoped claims.

8. Strengthen The Phase 6 Mechanism Claim

Goal: make the phase-6 replay explanation harder to dismiss as a post hoc interpretation.

- [] Check whether archive descriptor diversity predicts transfer after controlling for archive hardness and archive size bias.

- [] Verify that the residual-difficulty estimator is stable across baseline portfolios, not only one fixed reference set.
- [] Test whether the phase-6 mechanism claim holds if the train subset or synthetic family mix changes modestly.
- [] Add sensitivity analyses for archive size and replay-instance count.
- [] Report whether the diversity signal is driven by one descriptor family or remains broad-based.

Why this matters:

- "Random replay won" is interesting, but the mechanism claim still needs to survive basic robustness checks.

9. Tighten Phase 7 Discovery Validation

Goal: reduce the chance that a phase-7 "discovered operator" is only a renamed standard trick or a host-scaffold artifact.

- [] Add a manual literature audit for any operator that survives the automated filter.
- [] Expand transplant validation from held-out TSPLIB to a second scaffold pool if runtime allows.
- [] Check whether operator signatures remain stable when validation-family seeds are changed.
- [] Compare rediscovery counts using both exact signatures and coarser novelty-classification families.
- [] Add a blinded review step for the strongest surviving operator reports before making novelty claims.

Why this matters:

- The main scientific value of phase 7 is interpretability, so false-positive "discoveries" are especially costly.

10. Phase 8 High-Probability Closeout Phase

Goal: land one more bounded, positive result that is easier to defend than another open-ended operator-invention campaign.

- [x] Replace open-ended operator invention with feature-based selection or configuration over a fixed library of known operators and solver scaffolds.
- [x] Freeze a library of candidate low-level operators and host scaffolds before the study begins.
- [x] Reuse the phase-6 descriptor basis and add lightweight online search-state features for selector context.
- [x] Compare against the single-best fixed scaffold, full-solver evolution, and random operator selection.
- [x] Treat a statistically reliable held-out TSPLIB gain or a clean Pareto gain over the single-best fixed baseline as the success criterion in the protocol.
- [x] Keep the operator library interpretable enough that any win can be explained as selection, scheduling, or parameter control rather than hidden solver invention.

Research basis:

- Rice-style per-instance algorithm selection, as summarized in Algorithm selection on a meta level (<https://link.springer.com/article/10.1007/s10994-022-06161-4>), is a better fit when instance descriptors already exist and whole-solver discovery is unstable.
- Selection-oriented hyper-heuristics remain a standard, better-supported alternative to unconstrained heuristic generation; see Hyper-heuristics: A survey and taxonomy (<https://www.sciencedirect.com/science/article/pii/S0360835223008392>).
- Automatic configuration is already recommended practice in routing because it improves solution quality and supports fairer comparisons; see On automatic algorithm configuration of vehicle routing problem solvers (<https://link.springer.com/article/10.1007/s41604-019-00010-9>).

- A recent routing example using contextual low-level operator selection is Landscape-Aware Bandit Hyper-Heuristics for Online Operator Selection in UAV Inspection Routing (<https://arxiv.org/abs/2605.14620>).

Why this matters:

- The current evidence suggests the system is stronger at curation, scheduling, and tuning than at inventing wholly new reusable operators.
- That makes a selector/configurator phase the most plausible way to obtain a positive, interpretable claim without treating phase 7 as evidence of a reusable primitive.

Current status:

- Phase 8 is now implemented on branch adaptive-portfolio.
- The official evidence campaign is now complete on branch adaptive-portfolio.
- The adaptive controller beat the one-shot static LLM selector, but it did not beat the best fixed heuristic or the supervised selector on held-out TSPLIB.
- The oracle selector tied the best fixed heuristic on the primary held-out TSPLIB endpoint, so the current frozen portfolio has almost no single-best-versus-oracle headroom there.
- That means any further selector-focused work should first create portfolio complementarity or shift the main endpoint toward regimes where the oracle already improves over the single best heuristic.

11. Phase 9 Real-World Solver Evolution

Goal: test autonomous solver-code evolution on a real-world benchmark family with explicit feasibility logic, instead of another selector or operator-selection loop.

- [x] Pick one practical benchmark family with official public instances and best-known solutions.
- [x] Keep the first real-world phase on CVRP, not VRPTW, to reuse existing parsing/scoring infrastructure while still introducing real route-feasibility complexity.
- [x] Implement a strict parser, validator, and scorer for full returned solutions.
- [x] Define a bounded baseline set with 2-3 explicit solver heuristics.
- [x] Specify exactly what the LLM is allowed to evolve in code.
- [x] Match the bounded phase-9 validator to the unrestricted-route semantics used for the chosen Uchoa X instances, treating k as contextual metadata rather than a hard feasibility limit.
- [x] Curate the bounded phase-9 X subset so the committed train/holdout split includes both two-cluster and grid-like descriptor regimes, rather than only a single geometric mode.
- [x] Keep held-out evaluation frozen until the final solver is selected.
- [x] Run the official replicated phase-9 campaign.

Research basis:

- CVRPLIB is the standard public benchmark repository for CVRP instances and best-known solutions; see CVRPLIB (<https://galgos.inf.puc-rio.br/cvrplib/en/instances>).
- The Uchoa X set was designed to expose a wide range of structurally diverse CVRP behaviors and remains a standard benchmark for heuristic and exact evaluation; see Uchoa et al. 2017 (<https://doi.org/10.1016/j.ejor.2016.08.012>).
- DIMACS explicitly treats CVRP as a benchmark family where algorithm engineering, feasibility validation, and fair scoring matter; see DIMACS CVRP (<https://dimacs.rutgers.edu/programs/challenge/vrp/cvrp/>).

Why this matters:

- It is a stronger test of autonomous solver-code evolution than TSP portfolio selection.
- It keeps the claim bounded and scorable: progress toward feasible, competitive, interpretable solver logic on held-out CVRP.

Current status:

- Phase 9 is now evidence-complete on branch real-world-vrp.
- The official 20-offset campaign was technically valid: all runs completed, all solver-evolution final incumbents were feasible on both train and held-out panels, and the accepted path contained no fallback, generation-error, or solver-timeout contamination.
- Solver evolution beat the weaker nearest-neighbor and regret-insertion baselines on held-out CVRPLIB X instances, but it did not beat the strongest fixed Clarke-Wright baseline on the primary endpoint.
- Cross-family synthesis was completed from that evidence base, and the current narrow follow-up is a bounded phase-9 closeout study that disentangles direct synthesis, budget-matched search, and replay-aware iterative search without reopening the broader model-strength branch.

11A. Phase 9 Closeout: Budget-Control CVRP Study

Goal: answer the remaining mechanism question from phase 9 without changing benchmark family, validator, train/held-out split, or model tier.

- [x] Freeze a dedicated closeout protocol that keeps the bounded Uchoa X setup unchanged.
- [x] Add a `direct_generate_plus_one_repair` closeout arm.
- [x] Add a `budget_matched_no_replay` closeout arm with the same candidate budget as replay-aware search.
- [x] Add a `replay_solver_evolution` closeout arm that reuses the bounded phase-9 validation path.
- [x] Keep the original phase-9 fixed baselines available for contextual comparison.
- [x] Add a no-cost smoke config that exercises the full closeout condition set.
- [x] Run the official replicated closeout campaign and aggregate paired deltas against both the strongest fixed baseline and the budget-matched control.

Research basis:

- Budget matching is required before replay can be credited for anything beyond additional candidate budget; that is the same methodological point emphasized in the project's model-strength factorial protocol.
- Replicated run summaries, paired uncertainty intervals, and avoiding best-run-only interpretation follow the same empirical-discipline concerns raised in Deep Reinforcement Learning That Matters (<https://arxiv.org/abs/1709.06560>) and Deep RL at the Edge of the Statistical Precipice (<https://arxiv.org/abs/2108.13264>).

Current status:

- The implementation is on branch `cvrp-closeout-budget-control`.
- The protocol is `docs/PHASE_9_CLOSEOUT_BUDGET_CONTROL_CVRP_PROTOCOL.md`.
- The runbook is `configs/cvrp_phase9_closeout/RUNBOOK.md`.
- The official 20-offset campaign is now complete under `runs/cvrp_phase9_closeout/budget_control`, with aggregate results in `aggregate_20260605_050930_t`.
- `replay_solver_evolution` was the only learned arm that stayed fully feasible in all 20 official runs, and it beat both `budget_matched_no_replay` and `direct_generate_plus_one_repair` on held-out penalized gap.

- The closeout did not overturn the strongest fixed-baseline result: Clarke-Wright remained the best condition on the primary held-out endpoint.
- The permanent interpretation note is docs/PHASE_9_CLOSEOUT_BUDGET_CONTROL_CVRP_RESULTS_2026-06-05.md.

12. Cross-Family Code-Evolution Synthesis

Goal: turn the completed simple-game, TSP, and real-world CVRP evidence into a unified capabilities-and-limitations framework instead of adding another open-ended benchmark phase.

- [x] Normalize endpoint performance, code novelty, reliability, update/acceptance rate, and train-time trend signals across the completed official archives.
- [x] Keep train-time dynamics separate from held-out endpoint results.
- [x] Add a historical direct Codex-authored CVRP check that is evaluated through the phase-9 validator without API calls.
- [x] State the direct Codex result as descriptive and non-retained as thesis evidence unless a future protocol freezes and replicates direct-Codex prompting.
- [x] Preserve conservative wording: the synthesis supports capability-bounded adaptation, not a broad claim that LLM evolution beats mature optimization solvers.

Research basis:

- Replicated run summaries, uncertainty-aware reporting, and avoiding best-run-only interpretation follow the reproducibility concerns raised in Deep Reinforcement Learning That Matters (<https://arxiv.org/abs/1709.06560>) and Deep RL at the Edge of the Statistical Precipice (<https://arxiv.org/abs/2108.13264>).
- The selector and portfolio phases remain naturally connected to Rice-style algorithm selection; see The Algorithm Selection Problem ([https://doi.org/10.1016/S0065-2458\(08\)60520-3](https://doi.org/10.1016/S0065-2458(08)60520-3)).
- The CVRP real-world endpoint remains scoped to the standard CVRPLIB Uchoa X benchmark source used in phase 9; see CVRPLIB (<https://galgos.inf.puc-rio.br/cvrplib/en/instances>).

Current status:

- Cross-family synthesis is now implemented on branch cross-family-synthesis.
- The official synthesis artifact is under runs/cross_family_synthesis/meta_patterns_20260522_165500.
- The permanent report is docs/CROSS_FAMILY_CODE_EVOLUTION_SYNTHESIS_2026-05-22.md.
- The direct Codex CVRP solver was feasible on all phase-9 train and held-out instances and descriptively beat the fixed Clarke-Wright baseline on mean held-out penalized gap, but this remains a historical single-shot descriptive check rather than replicated stochastic evidence or retained thesis evidence.

13. Model Strength Versus Evolution Technique

Goal: separate base LLM coding strength from the added value of evolutionary code search, replay, and compressed failure memory.

- [x] Define a crossed design over simple games, symmetric TSP, and real-world CVRP.
- [x] Add exactly three model-strength tiers and record a transparent benchmark-strength score source.
- [x] Add the key budget_matched_no_replay control so replay is not credited for extra API budget alone.
- [x] Preserve fixed train/hold-out splits and existing validators.

- ☒ Aggregate into model x evolution matrices, within-task z-scores, variance decomposition, and effect sizes against both single-shot and budget-matched controls.
- ☐ Run the full paid campaign and interpret replay/failure/compression only when they beat budget_matched_no_replay.

Research basis:

- Stochastic adaptive systems need replicated seeds, uncertainty intervals, and effect sizes rather than best-run-only evidence; see Henderson et al. 2018, Deep Reinforcement Learning That Matters, and Agarwal et al. 2021, Deep RL at the Edge of the Statistical Precipice.
- Factorial designs and ANOVA-style variance decomposition are appropriate when the scientific question is whether one experimental factor, here base model strength, explains more variance than another factor, here evolution technique.
- Budget matching is required because otherwise replay and iterative evolution can be confounded with simply buying more candidate generations.

Current status:

- The implementation is on branch model-strength-factorial.
- The no-cost smoke artifacts are written under DO NOT COMMIT/.
- The official campaign should be run only after the smoke test passes and model access is confirmed.

Exit Condition

These items are reduced enough for a stronger publication push when:

- ☐ behavioral heuristics have human-validation evidence
- ☐ final evaluation is clearly separated from training-time selection
- ☐ main claims are replicated across multiple live runs
- ☐ qualitative case studies support the metric-based interpretation
- ☐ generalization claims are scoped to the tested benchmark regimes
- ☐ report language has been updated to reflect the stronger evidence